Rank order of odor-pair similarity, with toy simulations

> **QUESTION**
> Does blocking APL change the rank order of odor-pair similarity, in theory, in the model and in the recordings?

INPUTS

- The recordings (Report 06)
- The pinned model (Report 07)

OUTPUTS

- `odor_weighted/TNT_rank_order.csv`
- `odor_weighted/rank_order_toy_simulation.png`, `.json`, `_pairs.csv`

SCRIPTS

- `scripts/tnt_rank_order.py`
- `scripts/rank_order_toy_sim.py`
- `scripts/reports/r09_rank_order.py` (this report)

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Reports 06 and 07 look at how much each pair's similarity changes. A different question is whether the **order** of the pairs changes: if pair A is more similar than pair B with APL intact, is it still more similar with APL blocked? A purely global inhibition might be expected to keep the order. This report shows when it does and when it does not, then compares model and data.

Table 1. Symbols used in this report.

Symbol	What it stands for	Values, units, range
S^{on}_{ab}, S^{off}_{ab}	Cosine similarity of odors a, b with inhibition on (APL intact) and off (APL blocked)	−1 to 1
ρ	Spearman rank correlation over the odor pairs between S^{on} and S^{off}	1 = same order
τ	Kendall rank correlation over the same pairs	−1 to 1
swap	Two odor pairs whose order differs between on and off	
s_o	A divisor applied to every KC's response to odor o (case 1)	> 0
θ	KC threshold	
ρ_{within}, ρ_{between}	Spearman between two halves of the recordings: same condition, or TNT vs TNT3c	−1 to 1

1 How a change of order is measured

With 16 odors there are 120 pairs and 7,140 comparisons between two pairs. Spearman ρ correlates the ranks of S^{on} and S^{off} over the pairs. Kendall τ counts comparisons directly; without ties,

$$\tau = 1 - 2\,\frac{\#\,\text{swaps}}{\#\,\text{comparisons}}, \qquad \text{share swapped} = \frac{1-\tau}{2} \tag{1}$$

ρ = 1 means no swaps. ρ is less sensitive than τ to swaps between pairs with nearly equal similarity.

2 When does global inhibition keep the order?

- **Case 1: inhibition only scales.** If every KC's response to odor o is divided by the same number s_o (which may differ between odors), cosine similarity does not change at all, because cosine ignores the length of each response vector (Equation 2). The order is kept exactly.
- **Case 2: one threshold raise, odors alike.** If inhibition raises one threshold for every odor and KC, and every odor drives the KCs with the same statistics, then similarity depends only on how correlated the two odors' inputs are, through one rising curve at each threshold. The order is kept (up to sampling noise).
- **Case 3: odors differ.** If odors differ in strength (3a), the same raise removes a larger share of a weak odor's responders, so pairs of weak odors lose more similarity than pairs of strong ones. If some odors share their overlap mostly in a few strongly driven KCs (3b), those pairs keep more of their overlap. Either way some pairs swap.

$$\cos\left(\frac{\mathbf{y}_a}{s_a}, \frac{\mathbf{y}_b}{s_b}\right) = \frac{\mathbf{y}_a \cdot \mathbf{y}_b/(s_a s_b)}{\|\mathbf{y}_a\|\,\|\mathbf{y}_b\|/(s_a s_b)} = \cos(\mathbf{y}_a, \mathbf{y}_b) \tag{2}$$

3 Toy simulations of the three cases

No PN responses and no connectome: 1,000,000 model KCs each get a random input for each of 16 odors. Odors overlap through 4 shared random inputs (each odor takes its own mix) plus a private input. A KC's response is $\max(\text{input} - \theta, 0)$. With APL blocked 20% of KCs respond to an odor on average, and with APL intact 10%. In case 1 the divisor of each odor is $1 + aA_o$, with A_o the odor's mean response solved self-consistently, and one a for all odors set so the intact/blocked ratio averages 0.5.

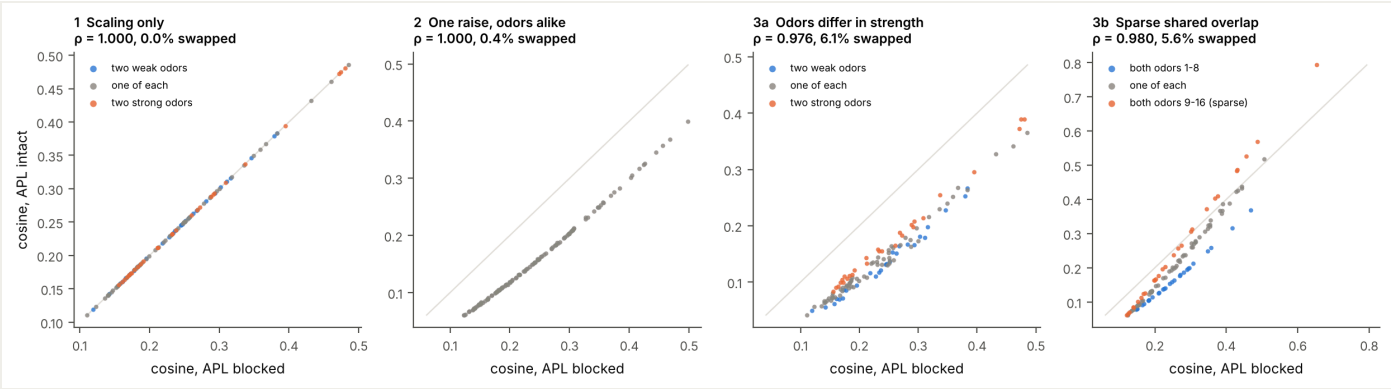


Figure 1. Rank order in the toy cases. Each point is one of the 120 odor pairs of a toy simulation. Horizontal: cosine similarity with APL blocked (20% of KCs respond per odor). Vertical: with APL intact (10%). The grey line marks equal values. A monotone cloud means the order is kept. Colors (where odors differ) show whether the pair has two weak odors, two strong ones, or one of each; in 3b, whether it shares the sparse overlap. Titles give Spearman ρ and the share of the 7,140 comparisons that swap.

- Scaling: every similarity unchanged to 10^{-14} (ρ = 1).
- One raise, odors alike: ρ = 0.9998; the 26 swaps are between pairs less than 0.003 apart (sampling noise).
- Odors differing in strength: ρ = 0.976 (6.1% swapped), and 0.956 (9.0%) with feedback instead of one raise.
- Sparse shared overlap: ρ = 0.980 (5.6% swapped).
- A divisor per KC instead of per odor keeps the order when it is random (ρ 0.9999) but not quite when it grows with the number of odors the KC responds to (ρ 0.995).

4 The connectome model

In the pinned model of Report 07, ρ is computed between the blocked and intact similarity over the odor pairs. Table 2 uses the similarity averaged over the three mushroom bodies first; Report 11 averages ρ over the mushroom bodies instead, which gives values about 0.005 lower.

Table 2. Rank order kept by every model version. Spearman over odor pairs between blocked and intact similarity of the pinned model, similarity averaged over hemibrain R and male CNS L and R.

version (Report 05 / 07)	ρ, ORN input (120 pairs)	ρ, GH146 input (78 pairs)
U	0.983	0.988
C	0.989	0.990
P0	0.984	0.989
P	0.953	0.960
L	0.972	0.982
M0	0.983	0.988
M	0.979	0.986
F0	0.984	0.983
F	0.984	0.984

Per mushroom body, uniform feedback gives 0.978, 0.981, 0.972 and a fixed threshold raise 0.980, 0.980, 0.974 (ORN input; hemibrain R, male CNS L, male CNS R). Only version P, with all inhibition dividing the PN bouton output per glomerulus, lowers ρ noticeably (0.95).

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5 The recordings

Over the 120 pairs, TNT and TNT3c similarity give ρ = 0.869, lower than any model version. Part of that is noise: even two sets of recordings of the same condition do not give ρ = 1. To separate the two, the recordings are split at random into halves (5 + 5 of the 11 TNT3c, 4 + 4 of the 8 TNT), 1,000 times. Two halves of the same condition give ρ_{within}; a TNT half against a TNT3c half gives ρ_{between}. If blocking APL kept the order, the two would be equal. Their ratio estimates the rank correlation without noise:

$$\rho_{\text{true}} \approx \frac{\rho_{\text{between}}}{\sqrt{\rho_{\text{within}}^{\text{TNT}}\ \rho_{\text{within}}^{\text{TNT3c}}}} \tag{3}$$

the usual correction for attenuation, applied to rank correlations of half-sample means

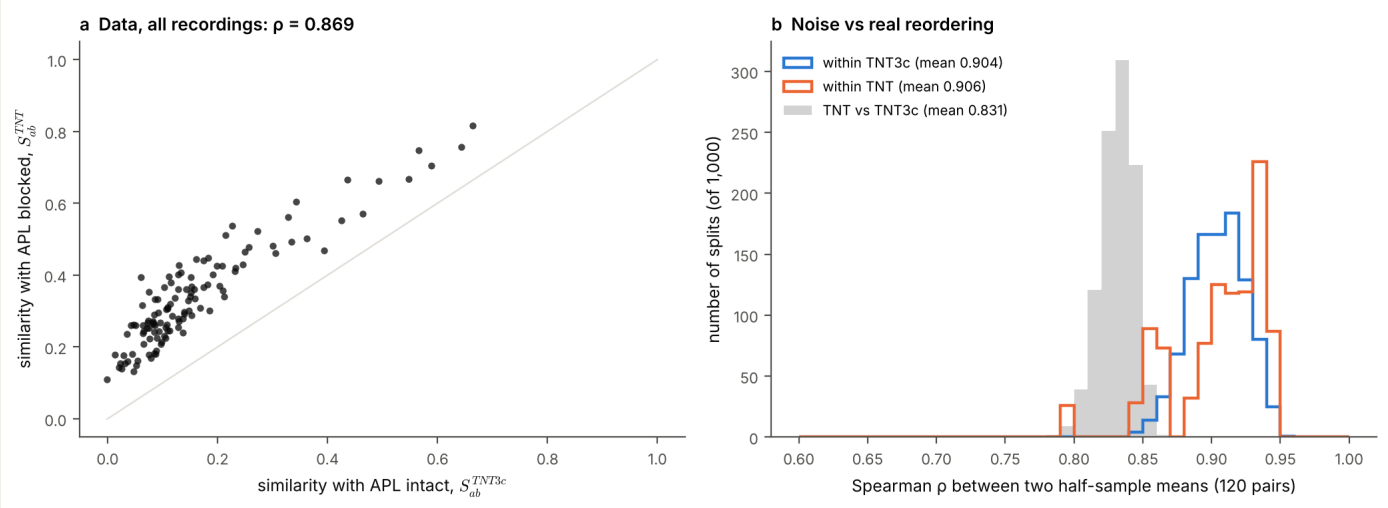


Figure 2. Rank order in the recordings. a, Each point is one odor pair: its mean similarity in TNT3c (horizontal) and TNT (vertical); the grey line marks equal values. All pairs lie above it, and the cloud is less tight than in the model. **b**, Distributions over 1,000 random splits of the recordings into halves: two halves of TNT3c (blue), two halves of TNT (orange), and a TNT half against a TNT3c half (grey). The between-condition values sit lower, so blocking APL changes the order more than noise does.

- ρ_{between} = 0.831 vs ρ_{within} = 0.904 (TNT3c) and 0.906 (TNT); the between value is lower than both within values in 97.5% of splits.
- Corrected for noise, ρ_{true} ≈ 0.92. The model versions give 0.95–0.99, so **the recordings reorder odor pairs more than uniform or connectome-weighted feedback inhibition on the KCs predicts**.
- Counting only responses above 1 (0/1) weakens the effect: 0.85 between vs 0.88 and 0.89 within, lower in 81% of splits.

What could reorder more

Inhibition that changes the input pattern differently for each odor, rather than one threshold per odor. In the model, presynaptic inhibition of PN boutons weighted per glomerulus (version P) does this, reaching ρ ≈ 0.95. Report 11 tests a local, non-spiking APL on KCs and boutons.

Differences between the recorded flies and the connectome flies, or response nonlinearities not in the model.

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